Personalized INCOSE Systems Engineering HANDBOOK Summary Notes

INCOSE SE Handbook	A GUIDE FOR SYSTEM LIFE CYCLE PROCESSES AND ACTIVITIES

CHAPTER ONE — SYSTEMS ENGINEERING INTRODUCTION

What Is Systems Engineering?	A transdisciplinary, integrative approach using systems thinking to realize, use, and retire engineered systems that are fit for purpose across the life cycle.
Why Is Systems Engineering Important?	Improves mission success and reduces cost/schedule risk by creating a shared vision and applying proven life-cycle methods, especially under complexity and compressed timelines.
System Concepts	Frames the SoI, environment, boundaries, hierarchy, states/modes, and architecture to describe how systems behave and interact internally and with external systems.
System Boundary and the System of Interest	Defines the line between the SoI and its environment, clarifying interfaces and architecture so requirements and behaviors are correctly allocated.
Emergence	Whole-system properties arise from interactions among elements and with the environment; architecture should reinforce desirable emergence and constrain the harmful.
Interfacing Systems, Interoperating Systems, and Enabling Systems	External systems may interface, interoperate operationally, or enable life-cycle activities; considering all prevents missed requirements and lifecycle pitfalls.
System Innovation Ecosystem	Enterprise learning model linking the Engineered System (System 1), Life-Cycle Management (System 2), and Enterprise Improvement (System 3) to drive innovation and adaptation.
The Hierarchy within a System	Systems decompose into elements (atomic or further systems); thoughtful partitioning (e.g., PBS, 7Â±2 span-of-control) balances manageability with integration needs.
Systems States and Modes	States are measurable conditions; modes are distinct operating capabilities; transitions are driven by events, decisions, and changes in state variables.
Complexity	Simple/complicated vs. complex systems demand tailored approaches; complex contexts need iterative, holistic methods beyond reductionist decomposition.
Systems Engineering Foundations	Foundational topics uncertainty, cognitive bias, principles, heuristics, and systems scienceâ€"underpin effective SE decisions and practice.
Uncertainty	Epistemic (knowledge) and aleatory (randomness) uncertainties pervade SE; reduce key knowledge gaps and record risk when deciding under uncertainty.
Cognitive Bias	Systematic judgment errors (e.g., anchoring, availability, groupthink) affect decisions; mitigate via independent reviews, structured methods, and disconfirming evidence.
Systems Engineering Principles	Transcendent guidance (15 principles) that inform SE across contexts and life-cycle stages, grounding why, what, and how we engineer systems.
Systems Engineering Heuristics	Practice-based "rules of thumb™ (e.g., model before build, question the stated problem) that capture hard-won wisdom when analytics alone fall short.
System Science and Systems Thinking	A systems worldview (General System Theory, cybernetics, complexity) provides concepts and methods; applying and reflecting in SE creates a virtuous improvement cycle.

CHAPTER TWO — SYSTEM LIFE CYCLE CONCEPTS, MODELS, AND PROCESSES

Life Cycle Terms and Concepts	Defines shared terms (life cycle, stages, artifacts) to manage systems from concept through retirement.
Life Cycle Characteristics	Highlights that life cycles are tailorable, risk-driven, and shaped by domain, constraints, and uncertainty.
Typical Life Cycle Stages	Common progression: Concept → Development → Production → Utilization → Support → Retirement.
Decision Gates	Formal go/no-go checkpoints where evidence is judged against criteria before committing to the next stage.
Technical Reviews and Audits	Structured evaluations (e.g., SRR/PDR/CDR, audits) that confirm readiness and compliance of products and processes.
Life Cycle Model Approaches	Families of ways to organize work across stages to balance risk, feedback, and delivery cadence.
Sequential Methods	Plan-driven, phase-to-phase flow with well-defined outputs before advancing (e.g., waterfall).
Incremental Methods	Deliver capability in planned increments, enabling early value and managed learning across releases.
Evolutionary Methods	Continuously evolve capability via rapid cycles and feedback to handle uncertainty and change.

System Life Cycle Processes	The standardized set of processes applied across the life cycle to achieve and sustain system outcomes.
Introduction to the System Life Cycle Processes	Explains purpose, scope, outcomes, and information items of each ISO 15288-aligned process group.
Concurrency, Iteration, and Recursion	Apply processes in parallel, revisit them as knowledge grows, and reuse patterns across levels of the system.
Agreement Processes	Buyer–supplier arrangements that establish, monitor, and fulfill agreements for products and services.
Acquisition Process	Defines needs, solicits solutions, selects suppliers, and manages delivery against contract objectives.
Supply Process	Plans and controls how a supplier delivers conforming results and evidence to the acquirer.
Organizational Project-Enabling Processes	Enterprise-level processes that enable projects with resources, methods, governance, and improvement.
Life Cycle Model Management Process	Defines and maintains the organization's life cycle models and tailoring guidance.
Infrastructure Management Process	Provides and sustains facilities, tools, environments, and services needed for engineering and operations.
Portfolio Management Process	Balances investments and priorities across projects to meet strategic and mission objectives.
Human Resource Management Process	Develops, assigns, and grows competencies to staff projects and sustain organizational capability.
Quality Management Process	Establishes quality policy and continuous improvement to meet stakeholder expectations across projects.
Knowledge Management Process	Captures, organizes, and reuses information and lessons to improve decisions and execution.
Technical Management Processes	Project-level processes that plan, monitor, control, and coordinate technical effort.
Project Planning Process	Establishes scope, strategy, plans, schedules, resources, and baselines for executing the work.
Project Assessment and Control Process	Measures progress, compares to plan, and takes corrective action to keep outcomes and baselines on track.
Decision Management Process	Uses criteria-based analyses to choose among alternatives and record the rationale.
Risk Management Process	Identifies, analyzes, treats, and monitors risks and opportunities across the life cycle.
Configuration Management Process	Maintains product and configuration baselines with controlled changes and traceability.
Information Management Process	Plans, controls, and protects information and communications to support collaboration and compliance.
Measurement Process	Defines measures, collects data, and analyzes results to inform decisions and improvement.
Quality Assurance Process	Assures processes and products conform to policies, standards, and contractual obligations.
Technical Processes	Technical activities that transform needs into solutions and sustain them in operation.
Business or Mission Analysis Process	Understands mission/business problem, context, and constraints to define capability needs.
Stakeholder Needs and Requirements Definition Process	Elicits and structures stakeholder expectations into clear, testable needs.
System Requirements Definition Process	Transforms stakeholder needs into technical requirements for the system and elements.
System Architecture Definition Process	Defines the system's structure, allocations, and interfaces to meet requirements.
Design Definition Process	Develops detailed solution definitions for system elements fit for realization and verification.
System Analysis Process	Performs trades, modeling, and analyses to justify choices and optimize against constraints.
Implementation Process	Realizes system elements (hardware, software, humans, procedures) according to design.
Integration Process	Assembles elements, resolves interfaces, and builds integrated configurations.
Verification Process	Provides objective evidence that requirements are met by items and the integrated system.
Transition Process	Prepares and deploys the system into its operational environment with users and enabling systems.
Validation Process	Provides evidence that the delivered system fulfills intended use and stakeholder needs in context.
Operation Process	Operates the system to deliver services and monitor performance within constraints.
Maintenance Process	Sustains capability by correcting defects, adapting to change, and preventing degradation.
Disposal Process	Retires the system safely, preserving records and handling data/material per policy.

CHAPTER THREE — LIFE CYCLE ANALYSES AND METHODS

Quality Characteristics and Approaches	Defines measurable quality attributes (e.g., safety, security, resilience, sustainability) and methods for achieving them across the life cycle.
Introduction to Quality Characteristics	Introduces key system quality characteristics and explains how trade studies balance them to meet stakeholder priorities.
Affordability Analysis	Evaluates total ownership cost versus capability to ensure economic viability throughout the life cycle.
Agility Engineering	Enables rapid adaptation and responsiveness to change while maintaining disciplined systems engineering practices.
Human Systems Integration	Integrates human performance, safety, and usability considerations to optimize mission effectiveness and reduce error.
Interoperability Analysis	Assesses and ensures that systems can exchange information and services effectively within operational contexts.
Logistics Engineering	Designs and manages support systems to ensure availability of materials, spares, and services throughout operation and sustainment.
Manufacturability and Productivity Analysis	Analyzes producibility and manufacturing efficiency to reduce cost, improve throughput, and ensure consistent quality.
Reliability, Availability, Maintainability Engineering	Ensures the system performs dependably and can be maintained and restored efficiently to meet mission needs.
Resilience Engineering	Designs systems that anticipate, absorb, and recover from disruptions while continuing to achieve critical objectives.
Sustainability Engineering	Balances present needs with long-term environmental, economic, and social impacts to sustain operations and resources.
System Safety Engineering	Identifies and mitigates hazards to prevent loss of life, mission, or assets through proactive risk analysis and control.
System Security Engineering	Protects systems and information from threats by integrating security controls into engineering activities and architectures.
Loss-Driven Systems Engineering	Focuses on preventing unacceptable losses across safety, security, and mission assurance domains using a unified approach.
Systems Engineering Analyses and Methods	Describes analysis methods supporting SE decisions throughout the life cycle to optimize design and reduce uncertainty.
Modeling, Analysis, and Simulation	Uses models and simulations to predict system behavior, assess performance, and inform trade and design decisions.
Prototyping	Builds early representations or prototypes to validate concepts, requirements, and interfaces before full-scale implementation.
Traceability	Ensures requirements, design, and verification artifacts are linked to maintain alignment and accountability across life cycle stages.
Interface Management	Manages and verifies physical, functional, and logical interfaces to ensure interoperability and integration success.
Architecture Frameworks	Provides structures (e.g., DoDAF, TOGAF) that organize system views and artifacts for consistent architectural reasoning.
Patterns	Applies reusable solution templates to common system or design challenges to accelerate and standardize engineering outcomes.
Design Thinking	Uses iterative, human-centered design to innovate solutions that align with user needs and operational realities.
Biomimicry	Draws inspiration from biological systems to develop efficient, adaptive, and sustainable engineering designs.

CHAPTER FOUR — TAILORING AND APPLICATION CONSIDERATIONS

Tailoring Considerations	Adapts the generic SE process set and rigor to suit project scope, risk, complexity, and stakeholder needs.
SE Methodology and Approach Considerations	Selects SE methods and life cycle models appropriate to system domain, maturity, and organizational capability.
Model-Based Systems Engineering	Applies modeling to represent, analyze, and manage system data, relationships, and decisions throughout the life cycle.
Agile Systems Engineering	Uses iterative, collaborative cycles emphasizing rapid delivery, stakeholder feedback, and adaptability to change.

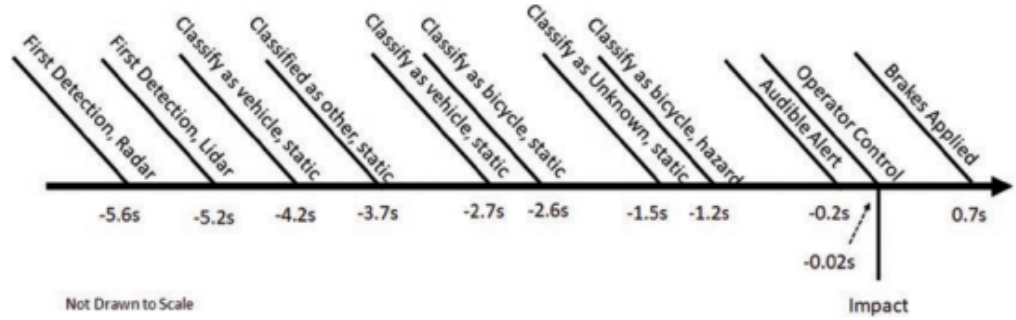
Lean Systems Engineering	Streamlines processes to eliminate waste and maximize value while maintaining discipline and traceability.
Product Line Engineering	Manages reusable assets, architectures, and components across product families to reduce cost and time-to-market.
System Types Considerations	Adjusts SE practices to the characteristics, constraints, and risks of the system domain or operational environment.
Greenfield or Clean Sheet Systems	Applies SE from a blank slate to develop entirely new systems or capabilities without legacy constraints.
Brownfield or Legacy Systems	Adapts SE to modernize, upgrade, or integrate existing systems with minimal disruption or rework.
Commercial Off-the-Shelf Based Systems	Integrates commercially available products into solutions, managing compatibility, configuration, and assurance gaps.
Software-Intensive Systems	Focuses SE on software architectures, code, and interfaces as dominant system components and risk drivers.
Cyber-Physical Systems	Coordinates engineering across tightly coupled physical and digital subsystems with real-time interactions.
Systems of Systems	Addresses large-scale, distributed systems whose elements are operationally and managerially independent.
Internet of Things and Big Data-Driven Systems	Applies SE to data-centric, connected systems emphasizing analytics, connectivity, and scalability.
Service Systems	Designs and manages systems that deliver ongoing functions or services rather than physical products.
Enterprise Systems	Applies SE to organizational-level systems integrating people, processes, information, and technology.
Application of Systems Engineering for Specific Product Sector or Domain Application	Provides domain-specific tailoring guidance for applying SE principles across diverse industries and missions.
Automotive Systems	Focuses on safety, reliability, and real-time control to meet stringent automotive regulatory and user requirements.
Biomedical and Healthcare Systems	Integrates regulatory compliance, safety, and usability to design systems supporting healthcare and life sciences.
Commercial Aerospace Systems	Balances innovation, safety, and certification rigor across complex, highly regulated aerospace environments.
Defense Systems	Implements SE to ensure mission assurance, survivability, and interoperability in defense and security systems.
Infrastructure Systems	Applies SE to complex civil systems where reliability, scalability, and stakeholder coordination are critical. (May reference systems reliant on SCADA/ICS)
Oil and Gas Systems	Manages SE for hazardous environments emphasizing safety, regulatory compliance, and continuous operation. (May reference systems reliant on SCADA/ICS)
Power and Energy Systems	Focuses on efficiency, reliability, and sustainability in energy generation, distribution, and storage systems. (May reference systems reliant on SCADA/ICS)
Space Systems	Ensures safety, reliability, and performance in high-risk space missions with rigorous verification and validation.
Telecommunication Systems	Coordinates SE for rapidly evolving communication networks emphasizing scalability and interoperability.
Transportation Systems	Applies SE to optimize mobility, safety, and infrastructure integration in multimodal transportation systems.

CHAPTER FIVE — SYSTEMS ENGINEERING IN PRACTICE

Systems Engineering Competencies	Defines the knowledge, skills, and behaviors required for effective SE practice across technical and leadership domains.
Difference between Hard and Soft Skills	Hard skills involve technical and analytical capabilities; soft skills enable collaboration, communication, and adaptability.
Systems Engineering Professional Competencies	Specifies competency areas such as technical processes, integration, analytical thinking, and lifecycle management mastery.
Technical Leadership	Combines domain expertise with influence, vision, and decision-making to guide multidisciplinary engineering teams.
Ethics	Upholds honesty, integrity, accountability, and respect for public safety and stakeholder trust in

	all engineering actions.
Diversity, Equity, and Inclusion	Promotes inclusivity, diverse perspectives, and equitable participation to enhance innovation and decision quality.
Systems Engineering Relationships to Other Disciplines	Explains how SE collaborates with related disciplines to integrate methods, optimize outcomes, and balance priorities.
Systems and Software Engineering	Coordinates system-level architectures and interfaces with detailed software development and integration activities.
Systems and Hardware Engineering	Aligns system architectures and design with hardware constraints, manufacturability, and reliability requirements.
Systems and Project Management	Integrates SE with project planning, scheduling, and resource control to balance scope, cost, and technical performance.
Systems and Industrial Engineering	Applies SE to optimize industrial workflows, production systems, and process efficiency within complex enterprises.
Systems and Operations Research	Uses analytical modeling, optimization, and quantitative analysis to support SE decision-making and trade studies.
Digital Engineering	Applies data-driven, model-based, and digitally integrated methods to enhance SE traceability and collaboration.
Systems Engineering Transformation	Describes the cultural and methodological shift toward agile, model-based, and data-centric SE practices enterprise-wide.
Future of Systems Engineering	Projects the evolution of SE toward increased automation, AI integration, and transdisciplinary system innovation.

CHAPTER SIX — CASE STUDIES

Radiation Therapy — The Therac-25	The Therac-25 tragedy highlights the catastrophic consequences of overreliance on software without sufficient redundancy or verification. AECL replaced proven hardware interlocks with software controls but failed to document, test, or validate safety-critical logic thoroughly. The result was lethal overdoses due to undetected software faults. This case underscores the need for rigorous system safety engineering, independent fail-safes, configuration control, and proper verification throughout the system life cycle—especially in life-critical systems.
Joining Two Countries — The Øresund Bridge	The Øresund Bridge success story demonstrates the power of disciplined systems engineering, stakeholder collaboration, and early risk management. From concept through construction, the project maintained strong governance, environmental stewardship, and cross-border cooperation. Engineers balanced national standards, safety, and environmental preservation through integrated planning, quantitative risk analysis, and inclusive stakeholder engagement—an exemplary model of SE applied to complex, multinational infrastructure systems.
Cybersecurity Considerations in Systems Engineering — The Stuxnet Attack	The Stuxnet cyber-attack on Iran's nuclear enrichment facility revealed how deeply cybersecurity must be integrated into systems engineering. The attack exploited zero-day vulnerabilities, supply chain weaknesses, and removable media to compromise an air-gapped industrial control system. It marked a turning point in cyber-physical threat sophistication, showing that SE must treat cybersecurity as a foundational element—considering insider threats, layered defenses, and continuous risk assessment across the system life cycle.
Design for Maintainability — Incubators	The incubator case illustrates how context-appropriate design and maintainability determine real-world system success. In developing regions, high-tech incubators often failed due to lack of spare parts and technical expertise. The NeoNurture project solved this by designing incubators built from automotive components available and repairable locally. This exemplifies SE's emphasis on lifecycle logistics, maintainability, and environmental suitability—proving that the best solution isn't always the most technologically advanced.
Artificial Intelligence in Systems Engineering — Autonomous Vehicles	 **FIGURE 6.1** Timeline of vehicle impact. From NTSB (2019a). Used with permission. All other rights reserved.

	The Uber autonomous vehicle fatality highlights the risks of inadequate integration between human operators and AI systems. Software misclassifications, delayed reactions, and disabled safety redundancies caused a preventable accident. Systems engineers must view autonomous systems as human-machine teams, designing for shared situational awareness, context-based decision-making, and robust safety management. This case drives home the importance of accountability, validation, and ethical considerations in AI-enabled systems.
Other Case Studies	Collectively, these case studies show that systems engineering principles—safety, collaboration, maintainability, and risk management—apply universally across domains. Whether preventing radiation overdoses, building bridges, defending against cyber threats, or designing AI systems, success depends on integrating human, technical, and environmental factors throughout the life cycle. SE provides the disciplined framework that turns complex, multidisciplinary challenges into sustainable, trustworthy systems.

CHAPTER SIX - APPENDICES

Helpful Terms & Definitions	
Ability — A natural or learned talent that lets someone do a task well.	**Measures of performance (MOPs)** — Indicators that the design meets performance requirements.
Acquirer — The stakeholder that buys or procures a product or service.	**N2 diagrams** — Grids that depict interfaces or relationships among system elements.
Activity — A cohesive set of tasks within a process.	**Need statement** — An agreed expression of what outcome or quality is required.
Agile systems-engineering — Applying an agile approach to the SE process.	**Operator** — See user.
Agile-systems engineering — Engineering that creates systems able to adapt quickly.	**Organization** — People and facilities arranged with roles and authorities.
Agreement — The mutually accepted terms that govern a working relationship.	**Performance** — Quantified attributes of how well, how much, how fast, or how ready.
Architect — See system architect.	**Physical configuration audit (PCA)** — Confirms the built system matches its configuration documentation.
Architecture — See system architecture.	**Process — Interrelated activities that transform inputs into outputs.**
Artifact — A work product used to capture and communicate project information.	**Product line** — A managed family of products with shared features for a market.
Attribute — An observable property or characteristic of a system or element.	**Project** — A time-bounded effort to deliver a defined product or service.
Baseline — The agreed snapshot of product attributes used to control change.	**Proof of concept** — A limited realization showing an idea is feasible.
Behavior — How a person or system acts or performs.	**Quality characteristics** — Inherent properties tied to requirements for product, process, or system.
Black box — External view of a system by its inputs/outputs and attributes only.	**Requirement statement** — A formal obligation for a function or quality the system must meet.
Brownfield SE — Engineering changes or additions within an existing legacy system context.	**Resource** — An asset used or consumed while executing a process.
Capability — What a system or process can achieve under stated conditions.	**Return on investment** — Benefit relative to development and production costs.
Commonality — Features shared across all products in a product line.	**Reuse** — Using an existing asset to solve a new problem.
Competence — The demonstrated ability to do something well.	**Skills** — Demonstrated ability to perform a learned physical or cognitive task.
Competency — The measurable skills, knowledge, and behaviors needed for a role.	**Stage** — A life cycle period tied to the system's state or realization.
Configuration item (CI) — A system element or artifact placed under configuration control.	**Stakeholder** — Anyone with needs, expectations, or claims on the system.
Customer — See acquirer.	**Supplier** — The party that provides a product or service under agreement.
Decision gate — An approval checkpoint with entry/exit criteria to proceed.	**System** — Interrelated elements that together exhibit behavior the parts alone do not.
Design constraints — The internal or external limits the solution must honor.	**System architect** — The role that defines and governs the system architecture and its realization.

Engineered system — A system designed for a purpose within given constraints.	**System architecture** — The fundamental concepts, properties, and governing principles of a system.
Enterprise — Interdependent resources organized to achieve business and operational goals.	**System element** — A constituent part of a system.
Environment — The surroundings in which the system is developed, used, and supported.	**System life cycle** — The system's evolution from conception through retirement.
Facility — Physical means or equipment that enable work to be performed.	**System of interest (SoI)** — The system whose life cycle is being engineered.
Failure — When a system or element does not meet its required performance.	**System of systems** — A system composed of independent systems working together.
Functional configuration audit (FCA) — Confirms the product meets its functional and performance baseline.	**Systems engineering** — A transdisciplinary approach to realize, use, and retire engineered systems.
Greenfield SE — Engineering a new system without significant internal legacy constraints.	**Tailoring** — Adapting documentation, processes, and reviews to the project's context.
Human factors — Applying knowledge about people to improve system performance and safety.	**Technical performance measures (TPMs)** — Metrics tracking key technical parameters and risks.
Interface — The defined boundary where two systems or elements interact.	**Trade-off** — Choosing among alternatives based on net stakeholder benefit.
IPO diagram — A high-level Inputs–Process–Outputs view of a process.	**User** — The person or organization that operates or benefits from the system.
Knowledge — Information applied to perform a function.	**Validation** — Evidence that the system fulfills intended use in its context.
Life cycle cost (LCC) — The total cost of a system from concept to retirement.	**Value** — The worth of an outcome relative to its cost.
Life cycle model — The framework of processes and activities across the life cycle.	**Variability** — Features that can differ among members of a product line.
Measure — A variable assigned a value from measurement.	**Verification** — Evidence that specified requirements have been met.
Measurement — The operations used to determine a value for a measure.	**Waste** — Work that does not add customer-perceived value.
Measures of effectiveness (MOEs) — Mission-level indicators of meeting stakeholder needs.	**White box** — Internal view of a system showing structure and attributes.

Input and Output Decisions			
Process Category	**Process**	**Typical Input / Output**	**Description**
Agreement	Supply	**Accepted system or system element**	System or element that the buyer formally accepts per the supply agreement.
Agreement	Acquisition	**Acquired system or system element**	System or element delivered to the buyer per the acquisition agreement.
Agreement	Acquisition	**Acquisition agreement**	The contract terms between buyer and supplier.
Agreement	Acquisition	**Acquisition need**	A need best met by buying from outside (or more economically supplied externally).
Agreement	Acquisition	**Acquisition payment**	Money or compensation paid for what was acquired.
Agreement	Acquisition	**Acquisition records/artifacts**	Saved documents and data about the acquisition.
Agreement	Acquisition	**Acquisition report**	Status/results summary of acquisition activities.
Agreement	Acquisition	**Acquisition strategy/approach**	How, when, with what resources the acquisition will be done.
Agreement	General Agreement	**Agreements**	All applicable agreements (e.g., acquisition and supply contracts).
Technical	Business or Mission Analysis	**Alternative solution classes**	The kinds of solution options that could address the need.
Technical	Business or Mission Analysis	**Analysis situations**	Analysis requests that arise from any stakeholder or process.

Org. Project-Enabling	Quality Management	**Applicable laws and regulations**	Laws and rules that must be followed.
Technical Management	Project Planning	**Breakdown structures**	Hierarchies (e.g., PBS/WBS/OBS) that organize work or product parts.
Technical	Business or Mission Analysis	**Business or mission analysis records/artifacts**	Saved data from business/mission analysis work.
Technical	Business or Mission Analysis	**Business or mission analysis report**	Status/results summary of business/mission analysis.
Technical	Business or Mission Analysis	**Business or mission analysis strategy/approach**	How business/mission analysis will be conducted.
Org. Project-Enabling	Portfolio Management	**Business plan**	The organization's overall business objectives and plan.
Technical Management	Configuration Management	**Candidate items for configuration management**	Things proposed to be placed under configuration control.
Technical Management	Information Management	**Candidate items for information management**	Things proposed to be placed under information control.
Technical Management	Risk Management	**Candidate risks and opportunities**	Newly identified risks/opportunities from anyone or any process.
Technical Management	Configuration Management	**Change request**	A formal request to fix, enhance, or alter something under control.
Technical	Business or Mission Analysis	**Concept of operations (ConOps)**	How leaders intend the organization to operate.
Technical Management	Configuration Management	**Configuration baseline**	The approved snapshot of configuration at a point in time.
Technical Management	Configuration Management	**Configuration management records/artifacts**	Saved data from configuration management activities.
Technical Management	Configuration Management	**Configuration management report**	Status/results summary of configuration management.
Technical Management	Configuration Management	**Configuration management strategy/approach**	How configuration management will be performed.
Technical Management	Configuration Management	**Configuration management system**	Tools/platforms used for configuration management.
Technical Management	Configuration Management	**Configuration verification and audit report**	Results of CM checks/audits for traceability and control.
Technical	Architecture Definition	**Constraints on solution**	Limits imposed on the system, its design, or the process.
Technical Management	Measurement	**Critical performance measurement data**	Collected data for the most important performance measures.
Technical Management	Measurement	**Critical performance measurement needs**	What decision makers need to measure most critically.
Technical	Validation	**Customer satisfaction inputs**	Feedback from surveys or other channels about customer satisfaction.
Technical Management	Decision Management	**Decision management records/artifacts**	Saved data from decision management work.
Technical Management	Decision Management	**Decision management report**	Status/results summary of decision management.
Technical Management	Decision Management	**Decision management strategy/approach**	How decisions will be prepared, made, and recorded.
Technical Management	Decision Management	**Decision register**	The log of decisions, context, rationale, and outcomes.
Technical Management	Decision Management	**Decision situations**	Decisions that need to be made (from any source/process).
Technical	Design Definition	**Design definition records/artifacts**	Saved data from design definition work.
Technical	Design Definition	**Design definition report**	Status/results summary of design definition.
Technical	Design Definition	**Design definition strategy/approach**	How the design will be defined and matured.
Technical	Disposal	**Disposal procedure**	Step-by-step instructions to dispose of the

			system.
Technical	Disposal	**Disposal records/artifacts**	Saved data from disposal activities.
Technical	Disposal	**Disposal report**	Status/results summary of disposal.
Technical	Disposal	**Disposal strategy/approach**	How disposal will be planned and executed.
Technical	Disposal	**Disposed system**	A system that's been deactivated, taken apart, and properly disposed of.
Technical	Architecture Definition	**Enabling systems**	External systems that support life-cycle activities (not part of operations).
Org. Project-Enabling	Human Resource Management	**Human resource management records/artifacts**	Saved data from HR management work.
Org. Project-Enabling	Human Resource Management	**Human resource management report**	Status/results summary of HR management.
Org. Project-Enabling	Human Resource Management	**Human resource management strategy/approach**	How staffing, skills, and growth will be handled.
Technical	Implementation	**Implementation records/artifacts**	Saved data from build/realization activities.
Technical	Implementation	**Implementation report**	Status/results summary of implementation.
Technical	Implementation	**Implementation strategy/approach**	How realization/build will be done.
Technical Management	Information Management	**Information management records/artifacts**	Saved data from information management.
Technical Management	Information Management	**Information management report**	Status/results summary of information management.
Technical Management	Information Management	**Information management strategy/approach**	How information will be governed and controlled.
Technical Management	Information Management	**Information register**	Repository for project information artifacts and access control.
Org. Project-Enabling	Infrastructure Management	**Infrastructure management records/artifacts**	Saved data from infrastructure management.
Org. Project-Enabling	Infrastructure Management	**Infrastructure management report**	Status/results summary of infrastructure management.
Org. Project-Enabling	Infrastructure Management	**Infrastructure management strategy/approach**	How facilities, tools, and services will be provided.
Technical	Transition	**Installation procedure**	Step-by-step instructions to install/transition the system.
Technical	Transition	**Installed system**	A system placed in its operational environment.
Technical	Integration	**Integrated system or system element**	A system or element created by combining constituent elements.
Technical	Integration	**Integration procedure**	Step-by-step instructions to integrate elements.
Technical	Integration	**Integration records/artifacts**	Saved data from integration activities.
Technical	Integration	**Integration report**	Status/results summary of integration.
Technical	Integration	**Integration strategy/approach**	How integration will be planned and executed.
Org. Project-Enabling	Knowledge Management	**Knowledge management records/artifacts**	Saved data from knowledge management.
Org. Project-Enabling	Knowledge Management	**Knowledge management report**	Status/results summary of knowledge management.
Org. Project-Enabling	Knowledge Management	**Knowledge management strategy/approach**	How knowledge is captured, shared, and reused.
Org. Project-Enabling	Knowledge Management	**Knowledge management system**	Tools/platforms used for knowledge management.
Org. Project-Enabling	Life Cycle Model Management	**Life cycle concepts**	Defined concepts for acquisition, operations, support, deployment, retirement, etc.
Org. Project-Enabling	Life Cycle Model	**Life cycle model management**	Saved data from life cycle model

	Management	**records/artifacts**	management.
Org. Project-Enabling	Life Cycle Model Management	**Life cycle model management report**	Status/results summary of LC model management.
Org. Project-Enabling	Life Cycle Model Management	**Life cycle model management strategy/approach**	How life cycle models and tailoring are managed.
Org. Project-Enabling	Life Cycle Model Management	**Life cycle models**	Frameworks organizing processes and stages across the life cycle.
Technical	Maintenance	**Maintained and sustained system**	A system kept operational via maintenance and support.
Technical	Maintenance	**Maintenance and logistics procedure**	Step-by-step instructions to conduct maintenance/logistics.
Technical	Maintenance	**Maintenance and logistics records/artifacts**	Saved data from maintenance/logistics.
Technical	Maintenance	**Maintenance and logistics report**	Status/results summary of maintenance/logistics.
Technical	Maintenance	**Maintenance and logistics strategy/approach**	How maintenance/logistics will be executed.
Technical Management	Measurement	**Measurement data**	Collected data for all measures (critical, organizational, project).
Technical Management	Measurement	**Measurement needs**	What needs to be measured (critical, organizational, project).
Technical Management	Measurement	**Measurement records/artifacts**	Saved data from measurement activities.
Technical Management	Measurement	**Measurement register**	Repository for measures, definitions, and status.
Technical Management	Measurement	**Measurement report**	Status/results summary of measurement.
Technical Management	Measurement	**Measurement strategy/approach**	How measures are defined, collected, and analyzed.
Technical	Operation	**Operation procedure**	Step-by-step instructions to operate the system.
Technical	Operation	**Operation records/artifacts**	Saved data from operations.
Technical	Operation	**Operation report**	Status/results summary of operations.
Technical	Operation	**Operation strategy/approach**	How operations will be conducted.
Technical	Operation	**Operational system**	A system in active use in its operational setting.
Org. Project-Enabling	Infrastructure Management	**Organization infrastructure**	Org-level facilities, people, resources, and services.
Org. Project-Enabling	Infrastructure Management	**Organization infrastructure needs**	What the organization requires to support work.
Org. Project-Enabling	Knowledge Management	**Organization lessons learned**	Organization-level lessons and improvements to retain.
Org. Project-Enabling	Quality Management	**Organization policies**	High-level organizational rules and direction.
Org. Project-Enabling	Portfolio Management	**Organization portfolio direction and constraints**	Org guidance and limits for the project portfolio.
Org. Project-Enabling	Quality Management	**Organization procedures**	Step-by-step instructions for org processes.
Org. Project-Enabling	Quality Management	**Organization processes**	The organization's defined processes.
Org. Project-Enabling	Quality Management	**Organization reports**	Compiled reports from org-level processes.
Org. Project-Enabling	Portfolio Management	**Organization strategic plan**	Mission/vision and strategic goals.
Org. Project-Enabling	Quality Management	**Organization strategies/approaches**	Strategies for org-level lifecycle processes.
Org. Project-Enabling	Life Cycle Model Management	**Organization tailoring strategy/approach**	How the org adopts/updates external standards.
Org. Project-Enabling	Measurement	**Org. measurement data**	Collected data for org-level measures.
Org. Project-Enabling	Measurement	**Org. measurement needs**	What org leaders need measured.
Technical	Validation	**Other validated artifacts**	Any artifacts that have been validated.

Technical	Verification	**Other verified artifacts**	Any artifacts that have been verified.
Org. Project-Enabling	Portfolio Management	**Portfolio management records/artifacts**	Saved data from portfolio management.
Org. Project-Enabling	Portfolio Management	**Portfolio management report**	Status/results summary of portfolio management.
Org. Project-Enabling	Portfolio Management	**Portfolio management strategy/approach**	How portfolio selection/balancing is done.
Technical	Business or Mission Analysis	**Problem or opportunity statement**	Concise description of the gap or new capability needed.
Technical Management	Project Assessment and Control	**Project assessment and control records/artifacts**	Saved data from monitoring/control.
Technical Management	Project Assessment and Control	**Project assessment and control strategy/approach**	How monitoring/control will be done.
Technical Management	Project Planning	**Project authorization**	Org. approval to proceed per plan.
Technical Management	Project Planning	**Project authorization request**	The project's request for approval to proceed.
Technical Management	Project Planning	**Project budget**	Cost estimate for the project (including reserves).
Technical Management	Project Planning	**Project constraints**	Limits imposed on the project.
Technical Management	Project Assessment and Control	**Project control request**	Direction to correct deviations from plan.
Technical Management	Decision Management	**Project decision gate/review result**	Artifacts/outcomes from gates or technical reviews.
Technical Management	Project Assessment and Control	**Project direction**	Org. guidance (continue, redirect, or end).
Org. Project-Enabling	Human Resource Management	**Project human resource needs**	People/skills required for the project.
Org. Project-Enabling	Infrastructure Management	**Project infrastructure**	Project-level facilities, tools, and services.
Org. Project-Enabling	Infrastructure Management	**Project infrastructure needs**	What the project needs to execute.
Org. Project-Enabling	Knowledge Management	**Project lessons learned**	Lessons and improvements specific to the project.
Technical Management	Measurement	**Project measurement data**	Collected data for project measures.
Technical Management	Measurement	**Project measurement needs**	What the project needs to measure.
Technical Management	Project Planning	**Project objectives**	The project's goals.
Technical Management	Project Planning	**Project planning records/artifacts**	Saved data from planning activities.
Org. Project-Enabling	Portfolio Management	**Project portfolio**	The set of projects aligned to strategy.
Technical Management	Project Planning	**Project procedures**	Step-by-step instructions for project technical activities.
Technical Management	Project Assessment and Control	**Project reports**	Compiled reports from project-level processes.
Technical Management	Project Planning	**Project schedule**	Linked activities/milestones with dates.
Technical Management	Project Assessment and Control	**Project status report/dashboard**	Current status, decisions, actions, and KPIs.
Technical Management	Project Planning	**Project strategies/approaches**	Strategies for all project-level processes.
Technical Management	Life Cycle Model Management	**Project tailoring strategy/approach**	How the project tailors life-cycle models.
Org. Project-Enabling	Human Resource Management	**Qualified personnel**	People authorized and competent to perform roles.
Technical Management	Quality Assurance	**Quality assurance corrective action**	Steps to fix or prevent recurrence of project nonconformities.
Technical Management	Quality Assurance	**Quality assurance evaluation report**	Results of QA evaluations and criteria used.
Technical Management	Quality Assurance	**Quality assurance records/artifacts**	Saved data from QA activities.
Technical Management	Quality Assurance	**Quality assurance report**	Status/results summary of QA.

Technical Management	Quality Assurance	**Quality assurance strategy/approach**	How QA will be planned and executed.
Technical Management	Quality Assurance	**Quality assurance system**	Tools/platforms used for QA.
Org. Project-Enabling	Quality Management	**Quality management corrective action**	Steps to fix or prevent org-level nonconformities.
Org. Project-Enabling	Quality Management	**Quality management criteria and methods**	Rules and methods used to judge/evaluate quality.
Org. Project-Enabling	Quality Management	**Quality management evaluation report**	Results of QM evaluations and criteria used.
Org. Project-Enabling	Quality Management	**Quality management records/artifacts**	Saved data from quality management.
Org. Project-Enabling	Quality Management	**Quality management report**	Status/results summary of quality management.
Org. Project-Enabling	Quality Management	**Quality management strategy/approach**	How quality is governed at org level.
Org. Project-Enabling	Quality Management	**Quality management system**	Tools/platforms used for quality management.
Technical Management	Information Management	**Records/artifacts**	The full set of saved evidence from all processes.
Agreement	Acquisition	**Request for supply**	The buyer's formal request to potential suppliers to bid/commit.
Technical	System Requirements Definition	**Requirements imposed on enabling systems**	Requirements that enabling systems must meet.
Technical	Implementation	**Reused system or system element**	A system/element used again to meet needs.
Technical Management	Risk Management	**Risk management records/artifacts**	Saved data from risk management.
Technical Management	Risk Management	**Risk management report**	Status/results summary of risk management.
Technical Management	Risk Management	**Risk management strategy/approach**	How risks/opportunities are managed.
Technical Management	Risk Management	**Risk register**	Repository of risks, owners, status, and actions.
Technical Management	Information Management	**Source documents**	External references relevant to the current stage.
Technical	Stakeholder Needs and Requirements Definition	**Stakeholder identification**	List of stakeholders with interests/needs.
Technical	Stakeholder Needs and Requirements Definition	**Stakeholder needs and requirements**	Structured collection of stakeholder expectations and context.
Technical	Stakeholder Needs and Requirements Definition	**Stakeholder needs and requirements definition records/artifacts**	Saved data from SN&R definition.
Technical	Stakeholder Needs and Requirements Definition	**Stakeholder needs and requirements definition report**	Status/results summary of SN&R definition.
Technical	Stakeholder Needs and Requirements Definition	**Stakeholder needs and requirements definition strategy/approach**	How SN&R will be elicited/structured.
Org. Project-Enabling	Life Cycle Model Management	**Standards**	This handbook and other applicable standards/specifications.
Agreement	Supply	**Supplied system or system element**	The system/element delivered by a supplier per the supply agreement.
Agreement	Supply	**Supply agreement**	The contract terms between supplier and buyer.
Agreement	Supply	**Supply payment**	Money or compensation paid for what is supplied.
Agreement	Supply	**Supply records/artifacts**	Saved data from supply activities.
Agreement	Supply	**Supply report**	Status/results summary of supply activities.
Agreement	Supply	**Supply response**	A supplier's bid/proposal and method to meet the request.

Agreement	Supply	Supply strategy/approach	How supply will be executed.
Technical	System Analysis	System analysis records/artifacts	Saved data from system analysis.
Technical	System Analysis	System analysis report	Status/results summary of system analysis.
Technical	System Analysis	System analysis request	A request to perform a system analysis.
Technical	System Analysis	System analysis strategy/approach	How analyses will be planned and done.
Technical	Architecture Definition	System architecture assessment report	Results of architecture evaluations and criteria used.
Technical	Architecture Definition	System architecture definition records/artifacts	Saved data from architecture definition.
Technical	Architecture Definition	System architecture definition report	Status/results summary of architecture definition.
Technical	Architecture Definition	System architecture definition strategy/approach	How architecture will be defined.
Technical	Architecture Definition	System architecture description	The architecture's purpose, qualities, constraints, decisions, and concerns.
Technical	Architecture Definition	System architecture rationale	Why architectural choices and allocations were made.
Technical	Design Definition	System design assessment report	Results of design evaluations and criteria used.
Technical	Design Definition	System design characteristics	Measurable attributes that define the design.
Technical	Design Definition	System design description	What the design is and how it works.
Technical	Design Definition	System design rationale	Why design choices and allocations were made.
Technical	Implementation	System element	A realized element that meets its element requirements.
Technical	Design Definition	System element description	Detailed definitions enabling design, implementation, and test.
Technical	Architecture Definition	System interface definition	Descriptions of interfaces between systems/elements.
Technical	System Requirements Definition	System requirements	Structured functional, performance, and constraint requirements.
Technical	System Requirements Definition	System requirements definition records/artifacts	Saved data from system requirements definition.
Technical	System Requirements Definition	System requirements definition report	Status/results summary of system requirements definition.
Technical	System Requirements Definition	System requirements definition strategy/approach	How system requirements will be defined.
Technical	Architecture Definition	System viewpoints, views, and models	Viewpoints, views, and models used to represent the architecture.
Technical Management	Project Planning	Systems engineering management plan (SEMP)	How SE processes/activities will be executed to deliver quality.
Technical Management	Life Cycle Model Management	Tailoring records/artifacts	Saved data from tailoring decisions.
Technical Management	Configuration Management	Traceability mapping	Links between artifacts (e.g., reqs, design, tests).
Org. Project-Enabling	Human Resource Management	Trained personnel	People trained to operate, maintain, or support the system.
Org. Project-Enabling	Human Resource Management	Training materials	Materials used to train personnel.
Technical	Transition	Transition records/artifacts	Saved data from transition activities.
Technical	Transition	Transition report	Status/results summary of transition.
Technical	Transition	Transition strategy/approach	How transition/deployment will be performed.
Technical	Validation	Validated stakeholder needs and requirements	Stakeholder needs/requirements confirmed as correct/complete.

Technical	Validation	**Validated system**	A system shown to meet intended use in context.
Technical	Validation	**Validated system architecture and design**	Architecture/design shown suitable for intended use.
Technical	Validation	**Validation criteria**	What will be measured, by whom, and where to validate.
Technical	Validation	**Validation procedure**	Step-by-step instructions to perform validation.
Technical	Validation	**Validation records/artifacts**	Saved data from validation.
Technical	Validation	**Validation report**	Status/results summary of validation.
Technical	Validation	**Validation strategy/approach**	How validation will be planned and executed.
Technical Management	Configuration Management	**Variance/deviation/waiver request**	Request to accept a departure from requirements.
Technical	Verification	**Verification criteria**	What will be measured, by whom, and where to verify.
Technical	Verification	**Verification procedure**	Step-by-step instructions to perform verification.
Technical	Verification	**Verification records/artifacts**	Saved data from verification.
Technical	Verification	**Verification report**	Status/results summary of verification.
Technical	Verification	**Verification strategy/approach**	How verification will be planned and executed.
Technical	Verification	**Verified system**	A system shown to meet its specified requirements.
Technical	Verification	**Verified system architecture and design**	Architecture/design shown to meet specified requirements.
Technical	Verification	**Verified system requirements**	Requirements shown to be satisfied.

PROCESS INDICATOR KEYWORDS TO WATCH OUT FOR

🧾 Agreement Processes	
Acquisition Process	acquisition, buyer–supplier, contract, procurement, acceptance, delivery, payment, records
Supply Process	supplier, deliverable, bid/proposal, supply agreement, delivery, response, compensation, verification

🏢 Organizational Project-Enabling Processes	
Life Cycle Model Management Process	life-cycle models, tailoring, phases, frameworks, concepts, governance, standards
Infrastructure Management Process	facilities, tools, IT services, resources, environment, provisioning, sustainment
Portfolio Management Process	portfolio, selection, prioritization, balancing, investment, alignment, reporting
HR Management Process	staffing, competencies, roles, qualifications, development, training, personnel
Quality Management Process	quality policy, organizational QA, continuous improvement, evaluation, metrics, audit
Knowledge Management Process	knowledge capture, sharing, lessons learned, reuse, repositories, KM systems

⚙️ Technical Management Processes	
Project Planning Process	project plan, scope, schedule, cost, resources, objectives, WBS, baselines
Project Assessment and Control Process	monitoring, control, performance, metrics, reviews, corrective action, reporting
Decision Management Process	decision analysis, trade-off, alternatives, criteria, rationale, records, outcomes
Risk Management Process	risk identification, assessment, mitigation, opportunity, register, strategy
Configuration Management Process	baselines, change control, CCB, variance/waiver, audit, traceability
Information Management Process	information storage, records, version control, access, repositories, documentation
Measurement Process	metrics, data collection, performance indicators, dashboards, analysis, feedback
Quality Assurance Process	QA plan, verification, audits, evaluations, compliance, corrective actions

🧠 Technical Processes	
Business or Mission Analysis Process	mission need, problem statement, objectives, constraints, feasibility, strategy
Stakeholder Needs and Requirements Definition Process	stakeholder identification, needs, elicitation, context, validation
System Requirements Definition Process	system requirements, functional/performance, constraints, allocation, specification
Architecture Definition Process	architecture views, structure, interfaces, rationale, trade-offs, models
Design Definition Process	design characteristics, decomposition, rationale, modeling, synthesis, documentation
System Analysis Process	analysis, modeling, trade studies, simulation, performance, alternatives
Implementation Process	realization, build, code, fabrication, integration readiness, verification
Integration Process	integration plan, interfaces, procedures, records, system assembly, verification
Verification Process	criteria, procedures, tests, verified requirements, evidence, reports
Transition Process	deployment, installation, acceptance, readiness, transition plan, records
Validation Process	intended use, validation plan, operational environment, test results, suitability
Operation Process	operations, usage, procedures, support, performance monitoring
Maintenance Process	maintenance, repair, sustainment, logistics, updates, patching
Disposal Process	decommission, disposal, data sanitization, recycling, termination, records